

Francesco Valentini - Curriculum Vitæ

Personal Details

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Address: Università di Pisa,
Dipartimento di Economia e Management,
Via Cosimo Ridolfi 10, 56124, Pisa, Italy

Current Position

Assistant Professor of Econometrics (ECON-05/A)
Department of Economics and Management, University of Pisa, Pisa, Italy
Starting date: October 10, 2024

Qualifications

National Scientific qualification as Associate Professor in the Italian higher education system, in the call 2023/2025 (Ministerial Decree n. 1796/2023) for the disciplinary field of 13/A5 - Econometrics with validity 10/06/2026 - 10/06/2038

Research Interests

Panel Data, Fixed Effects, Discrete outcomes, Network Econometrics, Applied Microeconometrics, Applied Time Series Analysis

Past Positions

- Junior Assistant Professor of Statistics (SECS-S/01)
Department of Economic and Social Sciences, Marche Polytechnic University, Ancona, Italy
January 1, 2023 - October 9, 2024
- Postdoctoral Fellow in Econometrics (SECS-P/05) on the Project:
“Panel data models and machine learning methods for the Green Innovators’ performance evaluation”
Tutor: Prof. Claudia Pigni
Department of Economic and Social Sciences, Marche Polytechnic University, Ancona, Italy
September 1, 2022 - December 31, 2022
- Postdoctoral Fellow in Econometrics (SECS-P/05)
Project: “Conditional inference for discrete panel data models”
Tutor: Prof. Claudia Pigni
Department of Economic and Social Sciences, Marche Polytechnic University, Ancona, Italy
June 1, 2020 - May 31, 2022

Education

19/03/2020 - Ph.D. in Economics, Marche Polytechnic University
Doctoral thesis: “Three Essays on the Conditional Inference Approach for Binary Panel Data Models”
Supervisors: Prof. Francesco Bartolucci and Prof. Claudia Pigni

24/10/2015 - Master's Degree in Economics and Finance, Marche Polytechnic University,
110/110 magna cum laude

Publications and Research Activities

– *Peer-Reviewed Articles*

2025 - Claudia Pigini, Alessandro Sterlacchini and Francesco Valentini. "Persistence of R&D intensities in the world's top investors in R&D", *Economics of Innovation and New Technology*,
<https://doi.org/10.1080/10438599.2025.2595205>

2025 - Michele Fratianni, Federico Giri, Riccardo Lucchetti and Francesco Valentini. "Monetization and the Fiscal Multiplier", *Oxford Bulletin of Economics and Statistics*,
<https://doi.org/10.1111/obes.12658>

2024 - Francesco Bartolucci, Claudia Pigini and Francesco Valentini. "MCMC Conditional Maximum Likelihood for the two-way fixed-effects logit", *Econometric Reviews*,
<https://doi.org/10.1080/07474938.2024.2339145>

2024 - Riccardo Lucchetti and Francesco Valentini. "Linear models with time-varying parameters in gretl: a comparison of different approaches", *Computational Statistics*,
<https://doi.org/10.1007/s00180-023-01452-3>

2023 - Riccardo Lucchetti and Francesco Valentini. "Kernel-based Time-Varying IV estimation: handle with care", *Empirical Economics*, <https://doi.org/10.1007/s00181-023-02450-6>

2022 - Francesco Bartolucci, Claudia Pigini and Francesco Valentini. "Testing for state dependence in the fixed-effects ordered logit model", *Economics Letters*,
<https://doi.org/10.1016/j.econlet.2022.110964>

2022 - Francesco Bartolucci, Claudia Pigini and Francesco Valentini. "Conditional inference and bias reduction for partial effects estimation of fixed-effects logit models", *Empirical Economics*,
<https://doi.org/10.1007/s00181-022-02313-6>

2021 - Francesco Bartolucci, Francesco Valentini and Claudia Pigini. "Recursive computation of the conditional probability function of the quadratic exponential model for binary panel data", *Computational Economics*, <https://doi.org/10.1007/s10614-021-10218-2>

2021 - Chiara Casoli, Luca Pedini and Francesco Valentini. "Do Covid-19 mobility restrictions affect economic uncertainty in Italy? Evidence from a SVAR approach", *Economics Bulletin*

– *Conference Proceedings*

2019 - Chiara Casoli, Luca Pedini and Francesco Valentini. "Spatial models in gretl: the SPM package" in *GRETl 2019. Proceedings of the International Conference on the Gnu Regression, Econometrics and Time-series Library*, DOI:10.6093/978-88-6887-057-7

– *Books*

2023 - Francesco Valentini, Claudia Pigini and Francesco Bartolucci. "Advances in maximum likelihood estimation of fixed-effects binary panel data models" in *Trends and Challenges for Categorical Data Analysis*, edited by Maria Kateri and Irini Moustaki, Springer,
https://doi.org/10.1007/978-3-031-31186-4_9

– **Work in Progress**

“Grouped fixed effects regularization for binary choice models” (joint with C. Pignini and A. Pionati)

“Specification testing with Grouped Fixed Effects” (joint with C. Pignini and A. Pionati)

“Testing for dependence in pairs of binary outcomes by an approximating logit model”

– **Software Components**

- cquad: Conditional Maximum Likelihood for Quadratic Exponential Models for Binary Panel Data.
R package
Francesco Bartolucci, Claudia Pignini and Francesco Valentini. 2020
<https://cran.r-project.org/package=cquad>
- cquadr: run the cquad R package in Stata.
STATA package
Francesco Bartolucci, Claudia Pignini and Francesco Valentini. 2020
<https://ideas.repec.org/c/boc/bocode/s458852.html> and <https://github.com/fravale/cquadr>
- FLS: Flexible Least Squares.
gretl function package
Riccardo “Jack” Lucchetti and Francesco Valentini. 2023
https://gretl.sourceforge.net/current_fnfiles/FLS.zip
- GFE: Grouped Fixed Effects for panel data
gretl function package
Riccardo “Jack” Lucchetti, Alessandro Pionati and Francesco Valentini. 2026
https://gretl.sourceforge.net/current_fnfiles/GFE.zip
- ketvals: Kernel-based Time-Varying Least Squares.
gretl Function Package
Riccardo “Jack” Lucchetti and Francesco Valentini. 2021
https://gretl.sourceforge.net/current_fnfiles/ketvals.zip
- spm: Spatial Regression Models.
gretl Function Package
Chiara Casoli, Luca Pedini and Francesco Valentini. 2019
https://gretl.sourceforge.net/current_fnfiles/spm.zip

– **Research projects**

- Member of the research project:
Next Generation EU. Project CUP: Project Code ECS00000041, CUP I33C22001330007;
Title: Innovation, digitalization and sustainability for the diffused economy in Central Italy – VITALITY
Period: January 1, 2023 - October 9, 2024

Refereeing Activity

Econometrics, Journal of Economic Behavior & Organization, Psychometrika (Theory and Methods),
SN Business & Economics, Statistical Methods and Applications

Talks/Presentations

- 2026 - 5th Italian Workshop of Econometrics and Empirical Economics, Ancona, Italy.
Presentation: *"Grouped fixed-effects AKM for employer-employee data with limited mobility"*
- 2026 - 5th Italian Workshop of Econometrics and Empirical Economics, Ancona, Italy.
Presentation: *"Grouped fixed effects regularization for binary choice models"*
- 2025 - Invited speaker at "Special Seminars InsIDE Lab", University of Insubria, Varese, Italy.
Presentation: *"Grouped fixed effects regularization for binary choice models"*
- 2025 - 11th Italian Congress of Econometrics and Empirical Economics, Palermo, Italy.
Presentation: *"Specification testing with Grouped Fixed Effects"*
- 2023 - The 28th International Panel Data Conference, Amsterdam, Netherlands.
Presentation: *"Specification testing with Grouped Fixed Effects"*
- 2023 - 8th gretl Conference, Gdańsk, Poland.
Presentation: *"Linear models with time-varying parameters in gretl: comparing different approaches"*
- 2023 - 10th Italian Congress of Econometrics and Empirical Economics, Cagliari, Italy.
Presentation: *"Monetization and the fiscal multiplier"*
- 2022 - The 27th International Panel Data Conference, Bertinoro, Italy.
Presentation: *"MCMC Conditional Maximum Likelihood for the two-way fixed-effects logit"*
- 2022 - Challenges for Categorical Data Analysis 2022, Perugia, Italy.
Presentation: *"Dynamic Ordered Logit with Fixed-Effects: testing for state dependence"*
- 2022 - 3rd Italian Workshop of Econometrics and Empirical Economics, Rimini, Italy.
Presentation: *"MCMC Conditional Maximum Likelihood for the two-way fixed-effects logit"*
- 2021 - 9th Italian Congress of Econometrics and Empirical Economics, Cagliari, Italy [Web].
Presentation: *"Partial effects estimation for fixed-effects logit panel data models"*
- 2019 - 7th SiDE Workshop for PhD students in Econometrics and Empirical Economics, Bertinoro, Italy.
Presentation: *"A conditional approach to testing for endogenous sample and self-selection in fixed-effects logit panel data models"*
- 2019 - 6th gretl Conference, Naples, Italy.
Presentation: *"Spatial models in gretl: the SPM package"*

Teaching

- 2025/2026 - Member of the PhD Board of "PhD Programme in Economics" of the Tuscan Universities, University of Siena
- 2025~2026- Lecturer of "Econometrics for Financial Markets",
Master in Risk Management, University of Pisa
- 2024/2025 ~ 2025/2026 - Lecturer of "Macroeconometrics",
Master's Degree in Economics, University of Pisa
- 2024/2025 ~ 2025/2026 - Lecturer of "Advanced Econometrics",
Master's Degree in Economics, University of Pisa

2024/25 - Lecturer of “Statistics for technological and experimental research”,
Bachelor’s Degree in Cardiocirculatory and Cardiovascular Perfusion techniques,
Faculty of Medicine and Surgery, Marche Polytechnic University

2023/2024 ~ 2024/25 - Lecturer of “Statistics for technological and experimental research”,
Bachelor’s Degree in Biomedical Laboratory Techniques, Faculty of Medicine and Surgery,
Marche Polytechnic University

2023/24 - Lecturer of “Econometric methods in practice”
(coordinator: Prof. Claudia Pigini), Ph.D. in Economics, Marche Polytechnic University

2019/20 ~ 2023/24 - Lecturer of “Microeconometrics in practice”
(coordinator: Prof. Matteo Picchio), Ph.D. in Economics, Marche Polytechnic University

2019/20 ~ 2022/23 - Lecturer of “Time Series Econometrics in practice”
(coordinator: Prof. Giulio Palomba), Ph.D. in Economics, Marche Polytechnic University

2020/21 ~ 2022/2023 - Teaching assistant: “Economia Politica I” (Microeconomics),
Bachelor’s Degree in Economics and Business, Marche Polytechnic University

2020 ~ 2021 - Teaching assistant: “Practice sessions in R”
(coordinators: Prof. Riccardo Lucchetti and Prof. Maria Cristina Recchioni)
Specialization Course in Data Science for Enterprises, Istituto Adriano Olivetti, Ancona, Italy

2019/20 - Associate Lecturer (Docente a Contratto) of “Economic and Financial Time Series Analysis”,
Master’s Degree in Finance and Quantitative Methods for Economics,
Department of Economics, University of Perugia, Perugia, Italy

2016/17 ~ 2018/19 - Tutor of “Economia Politica I” (Microeconomics),
Bachelor’s Degree in Economics and Business, Marche Polytechnic University

Memberships

2017-2019, 2021-2026 - SIde-IEA, *Italian Econometric Association*

Other Experiences

2025 - The 30th International Panel Data Conference, Montpellier, France

2024 - The 29th International Panel Data Conference, Orléans, France

02/2019 ~ 07/2019 - Visiting Ph.D. student at University of Perugia, courses attendance:

“Gaussian Mixture Models for Model Based Clustering and Classification”, Prof. Luca Scrucca

“Introduction to Latent Variable Models”, Prof. Francesco Bartolucci

2019 - 8th Italian Congress of Econometrics and Empirical Economics, Lecce, Italy

2018 - 1st Italian Workshop of Econometrics and Empirical Economics, Milan, Italy

Languages

Italian: Mothertongue

English: Upper Intermediate, Cambridge First Certificate in English (B2 Level)

German: Basic, basic words and phrases only

Computer Skills

Operating Systems: GNU/Linux, MacOS, Microsoft Windows

Text Editing and Office Skills: $\text{\LaTeX}$, LibreOffice, Microsoft Office

Software: R, gretl, STATA, MATLAB

Last update: September 3, 2026